Blackline Masters for Math II, Unit 1

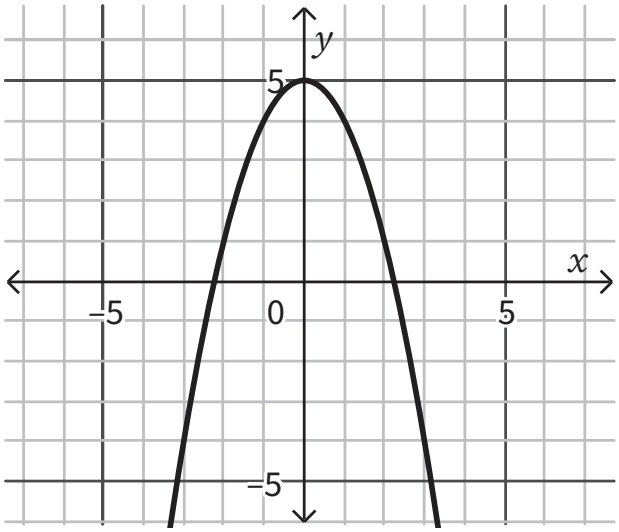
Lesson	Students per copy	Requires Cutting?	Written On?	Special Instructions
Lesson 6: Be the Change	2	Yes	Yes	Copy one set of cards for each pair of students in the class. Cut the cards and put them together to make a set that contains all the A, B, and C cards. Because students will be writing on the cards, they are not reusable.

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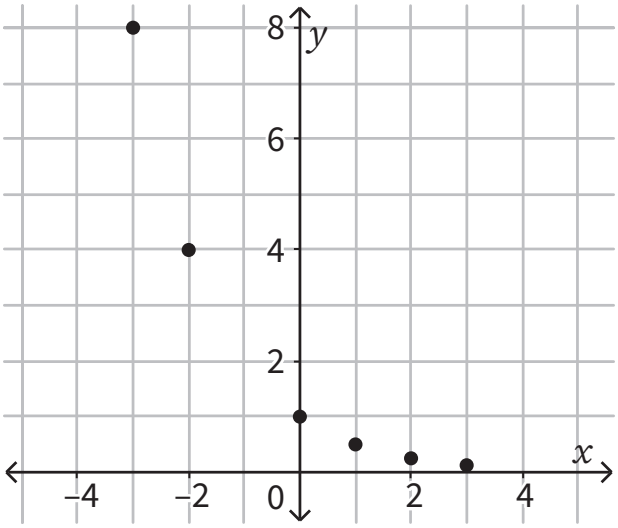
A1

The relationship between the total price of the service call and the number of hours worked for a plumber that charges a base fee of $\$55$ for a service call plus $\$35$ per hour for each hour worked during the service call.

A2



A3



A4

$$y = \frac{1}{3}(x-2)^2 + 4$$

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A5

$$y = \frac{1}{3}(x - 2) + 4$$

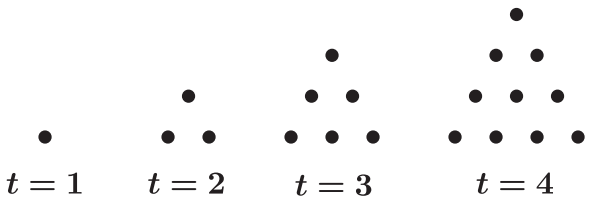
A6

The relationship between the speed of a car and the distance it takes to stop when traveling at that speed.

Speed (mph)	Stopping Distance (ft)
10	12.5
20	50
30	112.5
40	200
50	312.5
60	450

A7

The relationships between the number of dots in each figure and the time.



A8

The rate at which caffeine is eliminated from the bloodstream of an adult is about 15% per hour. The relationship between the amount of caffeine in the bloodstream and the number of hours from the time the adult drinks the caffeinated beverage if the initial amount of caffeine in the bloodstream is 500 mg.

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A9

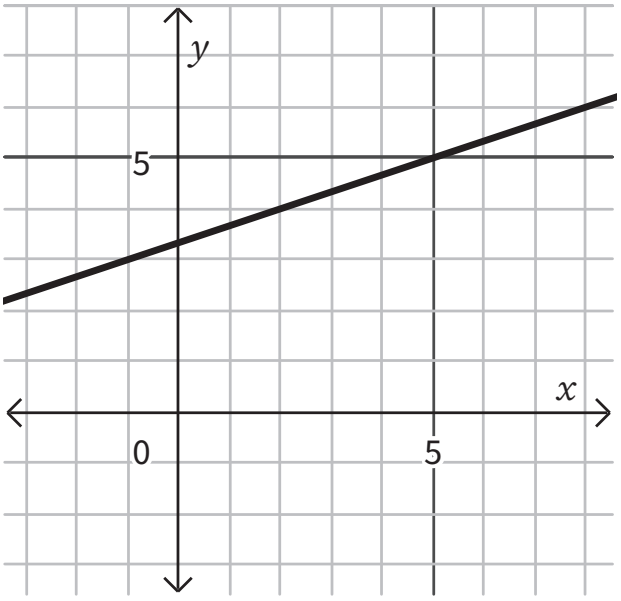
$$y = (4x + 3)(x - 6)$$

A10

Mary Contrary wants to build a rectangular flower garden that is 6 meters longer than it is wide. The function relates the area of the garden to its width.

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B1



B2

$$f(n) = \left(\frac{1}{2}\right)^n$$

B3

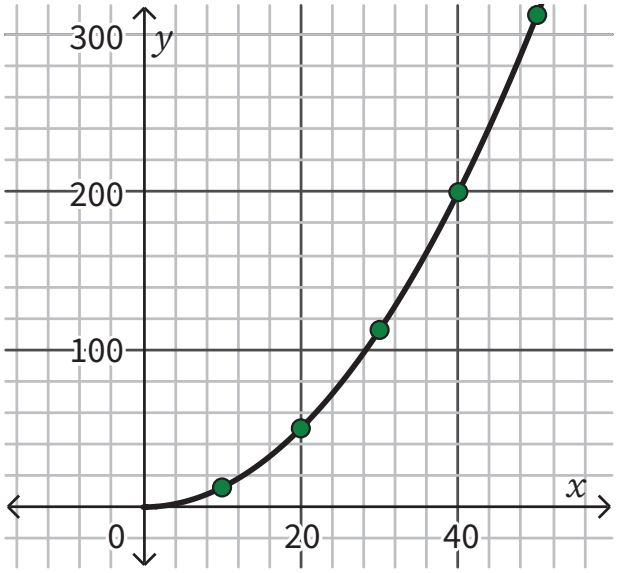
$$f(x) = x(x+6)$$

B4

x	$f(x)$	1st Diff
-1	4	$--$
0	5	1
1	4	-1
2	1	-3
3	-4	-5
4	-11	-7

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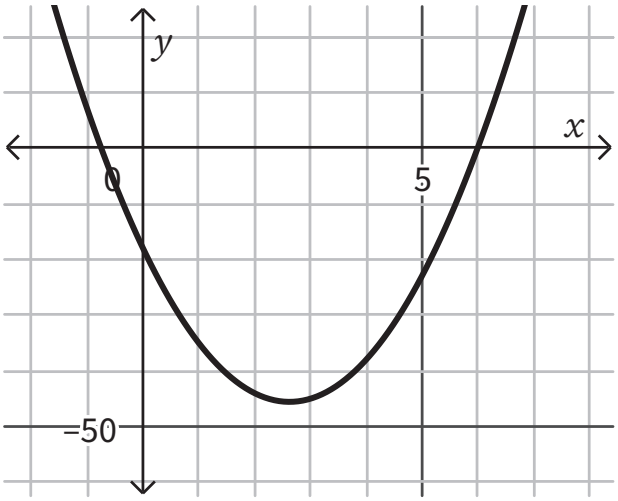
B5



B6

$$y = 55 + 35x$$

B7

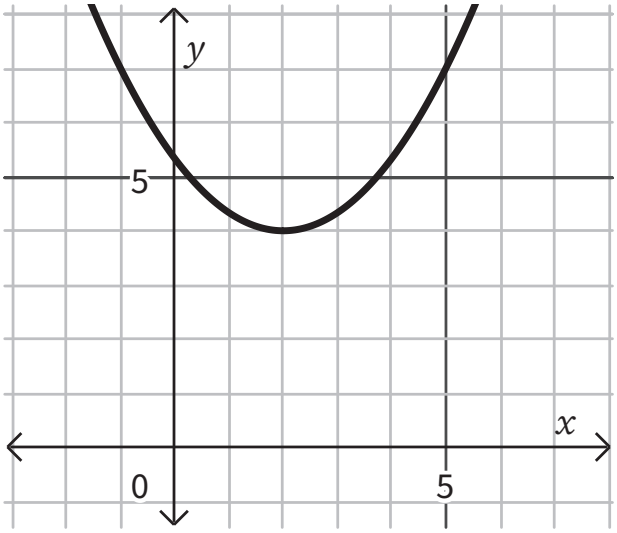


B8

x	$f(x)$
1	1
2	3
3	6
4	10
5	15

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B9



B10

$$f(x) = 500(.85)^x$$

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C1

Matches with _____

Type of function: _____

Type of change: _____

Second difference: _____

$312.5 < f(55) < 450$

C2

Matches with _____

Type of function: _____

Type of change: _____

Change factor for an interval of 1 is 0.85.

$f(0) =$_____ $f(3) =$_____

In this context, $f(3)$ means:

C3

Matches with _____

Quadratic function

Linear rate of change

x-intercepts: $\left(-\frac{3}{4}, 0\right)$ and _____

y-intercept:_____

C4

Matches with _____

Exponential function

Type of change: _____

Domain: $\{n | n \in Z, n \geq -3\}$

Range: _____

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C5

Matches with _____

Type of function: _____

Type of change: _____

Minimum: $(2, 4)$

Interval of decrease:

_____

Interval of increase: _____

C6

Matches with _____

Type of function: _____

Rate of change: _____

$f(2) =$_____

$f(x) = 6$ at $x =$_____

C7

Matches with _____

Linear function

Constant rate of change _____

Domain: _____

Range: $[55, \infty)$

C8

Matches with _____

Quadratic function

Type of change: _____

Second difference: _____

Domain: $\{n | n \in N\}$

Range: _____

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C9

Matches with _____

Quadratic function

Type of change: _____

$f(3) =$ _____

y-intercept: _____

C10

Matches with _____

Type of function: _____

Linear rate of change

Second difference: _____

Domain: $(-\infty, \infty)$

Range: _____

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